

Section A: Connected Dominating Set Solution

Problem: Connected Dominating Set (CDS) Decision Problem

Definition:

A dominating set D contains vertices such that every vertex outside D is adjacent to at least one vertex in D .

A connected dominating set additionally requires the induced subgraph of D to be connected.

Task 1 Solution:

A dominating set may not be connected. For example, in a graph containing two separated clusters, selecting one dominating vertex from each cluster creates a dominating set but not a connected one.

Every connected graph has a connected dominating set. A disconnected graph may not have one because the induced dominating vertices cannot be connected.

Task 2: CDS is in NP

Certificate:

A set of vertices D .

Verification:

1. Check that every vertex outside D has a neighbor in D .
2. Check that the subgraph induced by D is connected using BFS/DFS.

Both operations take polynomial time.

Task 3 and 4: NP-Hard Reduction

Reduce Vertex Cover to CDS.

Construction:

Given $G=(V,E)$ and k , create a graph where selecting vertices corresponding to a vertex cover produces a connected dominating set of bounded size.

Correctness:

If Vertex Cover has a solution of size k , the constructed CDS instance has a solution.

If the CDS instance has a solution, the corresponding vertices form a vertex cover.

The construction adds vertices and edges only polynomially, so the reduction is polynomial.